

**OSCILLOSCOPE DISPLAY CONTROL
34D
INSTRUCTION MANUAL**

NOVEMBER 1967

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INTRODUCTION

Digital Equipment Corporation of Maynard, Massachusetts, offers as part of its product line an oscilloscope display, designated Type 34D, which may be supplied as an option to a PDP-8 or PDP-8/S system. The display unit allows information from a digital computer to be displayed on the oscilloscope as a point of light whose location is specified by two 10-bit digital words.

The basic 34D Oscilloscope Display Unit consists of a Tektronix RM-503 Oscilloscope or equivalent and its associated logic controls.

A photomultiplier light pen, optional with the 34D, can be employed to exercise control over, or to communicate with, a running computer program.

SCOPE

This manual provides complete maintenance data on the logic circuitry, as it relates to the 34D Oscilloscope Display and both the PDP-8 and PDP-8/S. This manual also includes detailed information on logic operation, interface characteristics and connections, and installation and mounting.

OPERATION

The 34D Oscilloscope Display accepts both vertical and horizontal 10-bit coordinate address words from bits 2 through 11 of the PDP-8 accumulator. Pulses that are generated internally by the 34D Oscilloscope Display control logic transfer the 10-bit words into an X-axis buffer and a Y-axis buffer. The outputs of these buffers, converted from digital to analog levels, provide dc-level deflection inputs to the display oscilloscope. Logic circuits implement these function by interpreting commands (IOP pulses) generated in the computer.

A 6-bit address code, in the computer memory buffer, provides the basic activating signal to the logic control of the display. The sensing of the address code by the logic circuits enables the receipt of processor-generated IOP pulses. In the display control logic, IOT pulses control operation of the display unit. Upon receipt of IOP pulses, the functions of the IOT pulses that are generated in the display control logic are as follows.

IOT 051 (061) clears the 10-bit X (Y) buffer registers.

IOT 052 (062) loads the 10-bit X (Y) buffer registers with data from the processor accumulator.

IOT 054 (064) intensifies the point of light at the location determined by the X- and Y-address words.

IOT 071 allows the program to skip if the light pen flag is a ONE.

IOT 072 clears the light pen flag.

IOT 074 clears the brightness registers.

IOT 075 sets brightness register to 1.

IOT 076 sets brightness register to 2.

IOT 077 sets brightness register to 3.

3.1 Program Instructions

Table 3-1 lists the instructions from the processor that provide all necessary control over the operation of the Type 34D Oscilloscope Display.

Table 3-1
Control Instructions

Mnemonic	Octal Code	Operation
DCX (DCY)	6051 (6061)	Generates an IOP pulse at event time 1 to clear the X (Y) buffer in preparation for receiving new display data.
DXL (DYL)	6053 (6063)	Generates an IOP pulse at event time 1 (IOP1) and an IOP pulse at event time 2 (IOP2) to clear the X (Y) buffers and then load in new data from bits 2 through 11 of the accumulator.
DIX (DIY)	6054 (6064)	Generates an IOP pulse at event time 3 (IOP4) to intensify the point defined by the content of the X- and Y-coordinate buffers. This command can be combined with the DXL (DYL) command.
DXS	6057	Generates IOP pulses IOP1, IOP2, and IOP4 at event times 1, 2, and 3, respectively, to execute the combined functions performed by the DXL and DIX commands. The X-coordinate buffer is cleared and loaded with new data; then the point defined by the X- and Y-buffers is intensified.
DYS	6067	Generates IOP pulses IOP1, IOP2, and IOP4 at event times 1, 2, and 3, respectively, to execute the combined functions performed by the DYL and DIY commands. The Y-coordinate buffer is cleared and loaded with new data; then the point defined by the content of the X- and Y-buffers is intensified.
DSB	607X	Generates an IOP pulse at event time 3 (IOP4) to load the content of bits 10 and 11 of the instruction into the brightness register. When the instruction is 6075, the minimum brightness (0.4 μ s) is set, when 6076, the medium brightness is (0.8 μ s) is set, and when 6077, the maximum brightness (3 μ s) is set.

3.2 Logic Operation

The following paragraphs describe the operation of the logic control of the Type 34D Oscilloscope Display as it relates to the PDP-8 and PDP-8/S computers. Logic controls for the oscilloscope display appear on DEC engineering drawing BS-D-34D-0-2, sheets 1 and 2 located in section 8 of this manual. When referring to particular logic controls in the text, the sheet number and coordinate location will be given. The coordinate location is a letter/number designation used on DEC drawings to facilitate the location of individual logic controls.

3.2.1 Processor Turn-On - During the processor turn-on period, or whenever the START key is depressed, POWER CLEAR pulses (-3V, 100-ns pulses at a 10 KHz rate) clear the light pen flag flip-flop (sheet 1, D5) and set brightness registers 0 and 1 (sheet 1, B5) to the ONE state. With both brightness registers in the ONE state, the point of light on the oscilloscope is at maximum brightness.

3.2.2 Coordinate Channels - Because the horizontal (X) and vertical (Y) coordinate channels operate identically, only the horizontal channel will be discussed in detail.

3.2.2.1 Initiate - Computer-generated command pulses initiate operation of the 34D Oscilloscope Display. The computer sends the address code of the peripheral device over the I/O bus lines. This display control logic senses the address code, through connectors BOD3 and BOD4 (sheet 2, D1), in the W103 Device Selector at BOA10, BOB10 (sheet 2, C2, D2). (The address code is stored in computer memory buffer bits 3 through 8.) Sensing, by the display control logic, of its own discrete address code allows acceptance of the IOP pulses, generated by the computer, and provided at terminals K, M, and P of connector BOD2 (sheet 2, C1). The decoded address enables the input gates to the three pulse amplifiers (PA) on the W103 Device Selector at BOA10, BOB10 (sheet 2, C2). When enabled, these gates permit incoming IOP pulses to activate their respective PA to produce the IOT pulses actually used within the display control circuits.

3.2.2.2 Clearing and Loading the Buffer - Initially, the computer generates a DCX instruction, setting up the I/O address, and transmitting an IOP1 pulse to the display control through terminal K of connector BOD2 (sheet 2, C1). The 100-ns pulse (IOT 051), which is generated by the ADAE PA in the W103 Device Selector, clears all ten R203 Data Buffer Registers at connectors BOA16 through BOA18, BOA24 (sheet 2, B2 through B8). The DCX instruction is normally generated at the start of the program run to ensure that the data buffer registers are all in the ZERO state. For the remainder of the program, the clear instruction is combined with the load instruction (delayed by 1 μ s) to increase the speed of operation.

Typically, an IOP1 pulse (at event time 1) and an IOP2 pulse (at event time 2) are generated by the processor in the same computer cycle. The IOT 051 (event time 1) pulse, generated within this control logic, clears the ten R203 Data Buffer Registers (BOA16 through BOA18, BOA24) (sheet 2, B2 through B8) to prepare them for acceptance of new data. The IOT 052 pulse (positive-going transition at event time 2), occurring either 1 μ s or 4.75 μ s later, is applied to the ten DCD gates (sheet 2, B2 through B8) in the buffer. The instructions DXL and DCX (1 μ s and 4.75 μ s, respectively) determine when the IOT 052 pulse appears at the input to the DCD gates in the buffer.

The horizontal address word is loaded into the data buffer registers through connectors BOA1 and BOA2 (sheet 2, B2 through B8). This, together, with the positive-going IOT 052 pulse enables the DCD gate. When enabled, the DCD gate activates the data flip-flop register and sets it to the ONE level (-3V). With any bit a logic ZERO (-3V) at connectors BOA1 and BOA2, the corresponding DCD gate is inhibited preventing the data flip-flop from being set to the ONE state. Thus the output of the flip-flop is now set to the same number held by the accumulator.

3.2.2.3 Digital-to-Analog Conversion - The 10-bit horizontal address word at the output of the data flip-flops is converted into an analog voltage between ground and -10V. This is accomplished by four 3-input A601 Digital-to-Analog Converters at connectors BOA20 through BOA23 (sheet 2, B2 through B8). (The D/A converter, at connector BOA23, utilizes only the most significant input, pin U.) The analog voltage varies between its upper and lower limits in 1024 discrete intervals, and is directly proportional to the numerical value of the coordinate address.

A -3V input signal at all inputs to the D/A converter produces a ground level output. Conversely, a 0V at all inputs produces a -10V analog output (see Table 3-2). This output connects to the oscilloscope through pin D of connector BOA32 (sheet 2, A2). Ten data flip-flop registers provide accessibility to 1024 possible locations (2^{10}) on the oscilloscope.

Table 3-2
Coordinate-Channel Logic Conversion Table

Digital Address and Flip-Flop States	Buffer Output and D/A Converter Input	D/A Converter Output
1 1 1 1 1 1 1 1 1 1	x x x x x x x x x x	most positive (ground)
1 1 1 1 1 1 1 1 1 0	x x x x x x x x x 0	
.	.	
.	.	
.	.	
1 0 0 0 0 0 0 0 0 1	x 0 0 0 0 0 0 0 0 x	.

Table 3-2
Coordinate-Channel Logic Conversion Table (continued)

Digital Address and Flip-Flop States	Buffer Output and D/A Converter Input	D/A Converter Output
1 0 0 0 0 0 0 0 0 0	x 0 0 0 0 0 0 0 0 0	.
0 1 1 1 1 1 1 1 1 1	0 x x x x x x x x x	.
0 1 1 1 1 1 1 1 1 0	0 x x x x x x x x 0	.
.	.	.
.	.	.
.	.	.
0 0 0 0 0 0 0 0 0 1	0 0 0 0 0 0 0 0 0 x	.
0 0 0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0 0 0	most negative (-10volts)
0 = logic zero = ZERO state	0 = ground x = -3V	
1 = logic one = ONE state		

3.2.3 Intensity Circuits - At event time 3, the computer generates a DIX instruction and transmits an IOP4 pulse which is received by the 34D Oscilloscope Display at pin P of connector BOD2 (sheet 2, C1). The W103 Device Selector Pulse Amplifier (AR AS) generates a 100-ns pulse (IOT 054) which, in its positive-going transition, enables the EF DCD gate in the R302 at connector BOA29 (sheet 2, C2). This triggers the input to the delay and causes its output to change from a ground level (0V) to -3V for 1 μ s. When the output returns to ground, it is applied to and half-enables the three DCD gates controlling the high-, medium-, and low-intensity delay circuits. The level of brightness intensity is dependent upon the output of the two R202 Brightness Register Flip-Flops at connector BOA26 (sheet 1, B5). By way of example, the low brightness delay (sheet 2, C5) will operate only when the outputs from BR0 and BR1 are 0 and 1, respectively. When this occurs, the requirements for the DEJH diode gate (sheet 2, D4) are satisfied and the -3V level at inputs D and E is inverted by the JH gate to produce a ground level. This ground level is applied to the EF DCD gate (sheet 2, C4) and, when combined with the IOT 054 positive-going pulse, enables that gate. The DCD gate in turn triggers the JKM delay whose output is applied to the W681 Intensity Amplifier (INT AMP, sheet 2, C6) for 0.4 μ s. An output of 25V, produced by the intensity amplifier, is connected to the oscilloscope through pin F of connector BOA32 (sheet 2, C7). The medium and high brightness outputs are produced in the same manner as the low brightness output.

The brightness register is loaded from the contents of memory buffer bits 10 and 11 through connector BOA4 (sheet 1, C5) as a DSB instruction is generated by the PDP-8. If the instruction code is 6075, memory buffer bits 10 and 11 contain a 0 (-3V) and a 1 (GRD) which appear at pins T and V, respectively, of connector BOA4. The 0 level (-3V) at pin T is first inverted by the JK diode gate (sheet 1, C5) which provides a ground level to the DE DCD gate (sheet 1, B5). When coupled with an IOT 074 negative-going pulse, the DE DCD gate will be enabled placing brightness register BR0 in the CLEAR state. The 0 level (-3V) at pin T is also present at the level input to the KL DCD gate, but it only serves to inhibit the gate.

The 1 level (GRD) at pin V is inverted by the LM gate (sheet 1, C5) producing a -3V level output which inhibits the NP DCD gate (sheet 1, C5). The 1 level (GRD) at pin V half enables the UV DCD gate. An IOT 074 pulse fulfills the DCD gate, setting brightness register BR1 to the 1 state. (The two diode gates in R107 at connector BOA25 (sheet 1, B5) permit collector-triggering of the brightness register flip-flops by a power clear pulse.)

3.2.4 Light Pen Operation - A photomultiplier light pen may be used with the 34D to communicate with a computer program. Basically, the light pen detects a point of light on the CRT and supplies it as a negative voltage level to the display control circuits through pin V of connector BOA32 (sheet 1, D8). Because this signal is not a standard DEC level, it is buffered by an emitter follower in module W681 at connector BOA31 (sheet 1, D7). This buffering provides impedance and amplitude characteristics required for DEC logic operation. Pin R of the diode gate in module R111 at connector BOA27 (sheet 1, D6) receives the buffered signal and, together with the -3V light pen strobe at pin S is inverted by the transistor at pin U. This inverted signal collector triggers the light pen flag flip-flop (sheet 1, C5) setting it to the ONE state. The light pen strobe is generated by the W681 Intensity Amplifier at connector BOA31 (sheet 2, C6). The outputs of the 0.4, 0.8, and 3 μ s delays are OR-combined so that the activation of any one produces a strobe pulse of the same duration as the intensify signal that is transmitted to the oscilloscope.

The -3V output of the light pen flag flip-flop is inverted by the R111 at connector BOA27 (sheet 1, C6) and activated the interrupt line (pin M of BOA6, sheet 1, C7).

A ONE on the light pen flag flip-flop also half-enables the skip command input gate (sheet 1, pin E, C6). The requirements for a skip are completed when bit 9 of the memory buffer register contains a ZERO (pin D) and when an IOT 071 signal is generated (pin F). The diode gate connected to the three conditions acts as a NAND gate driving an inverter. Thus when all three inputs are ONES, the output at pin H is at a GND level. This GND level is brought out to pin K of connector BOA6 (sheet 1, C6) to initiate the skip instruction.

The light pen flag flip-flop can be cleared by a power-clear pulse or by the combination of an IOT 072 pulse and bit 9 of the memory buffer in the ZERO state. The IOT 072 pulse is generated by the AK AL power amplifier in the W103 Device Selector at connector BOA12/BOB12 (sheet 1, B3). It is applied as a ground level to the UV DCD gate in module R203 at connector BOA24 (sheet 1, D5). Bit 9 of the memory buffer at pin S of connector BOA4 is a -3V level which is inverted and applied to the UV DCD gate. With pins U and V qualified, the DCD gate is enabled and clears the light pen flag flip-flop.

4 INTERFACE

This section illustrates the interface connections and signals between the PDP-8 and the PDP-8/S and the 34D Oscilloscope Display Unit. Figures 4-1 and 4-2 show the interface between the PDP-8 and the 34D. Figures 4-3 through 4-5 illustrate the interface connections between the PDP-8/S and 34D logic control. Figure 4-6 indicates the connections between the 34D logic and the oscilloscope.

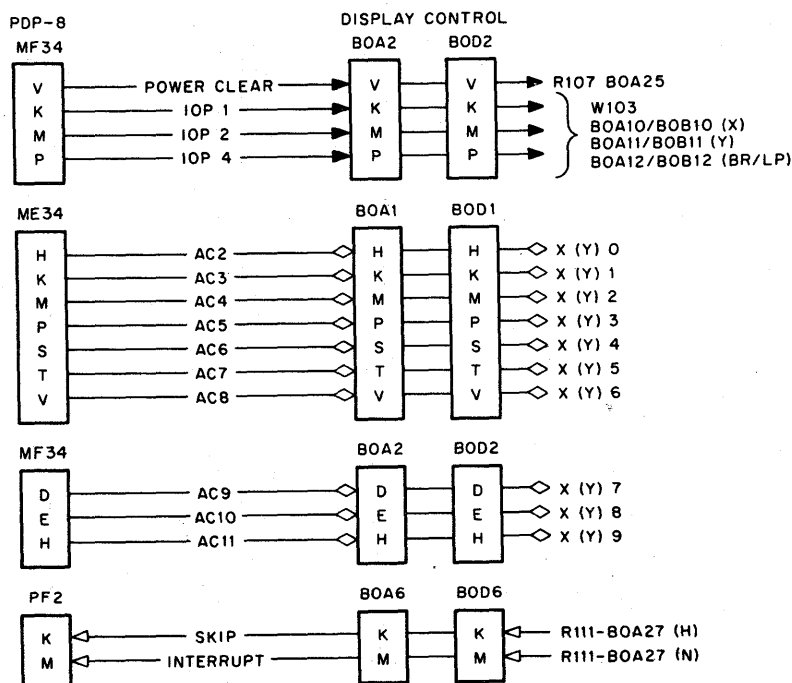


Figure 4-1 PDP-8 to 34D Control Interface, X/Y Address Data and Function Control Signals

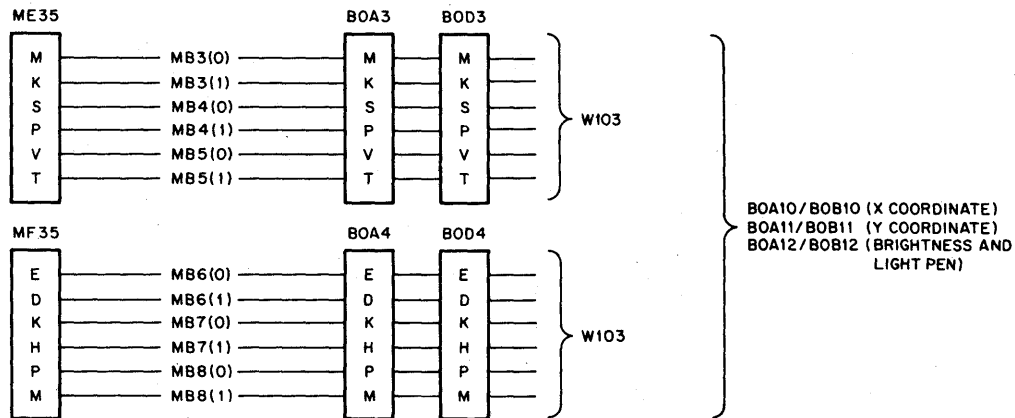


Figure 4-2 PDP-8 to 34D Control Interface, X and Y Coordinate Address and Brightness and Light Pen Address

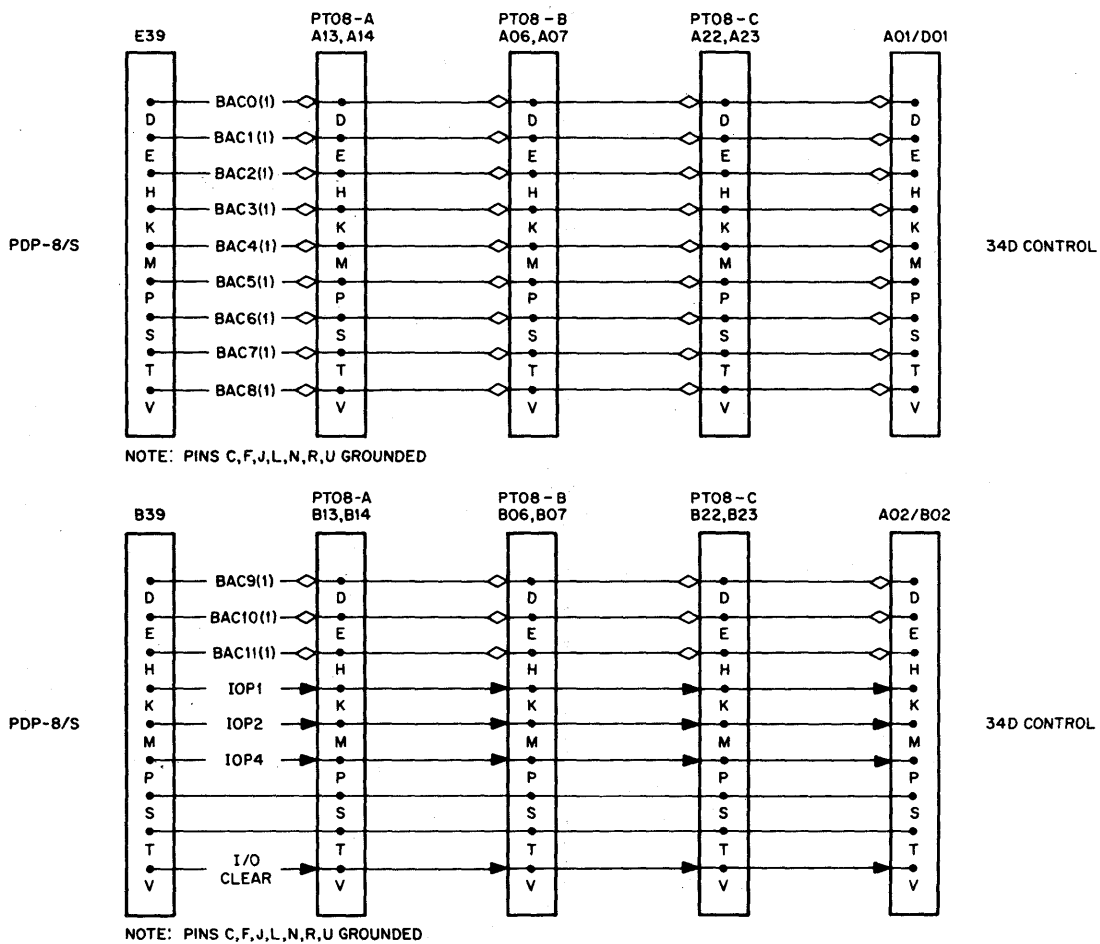


Figure 4-3 PDP-8/S to 34D Control Logic Interface

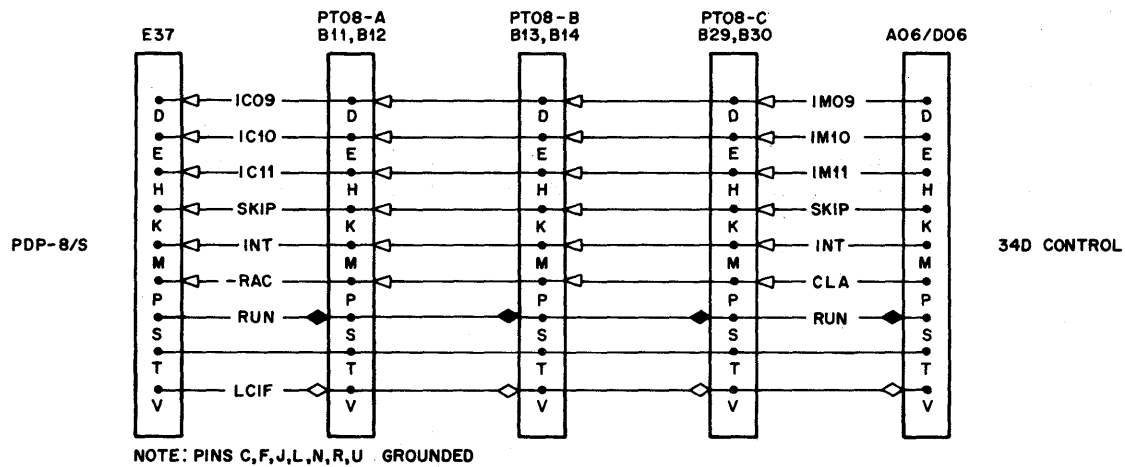
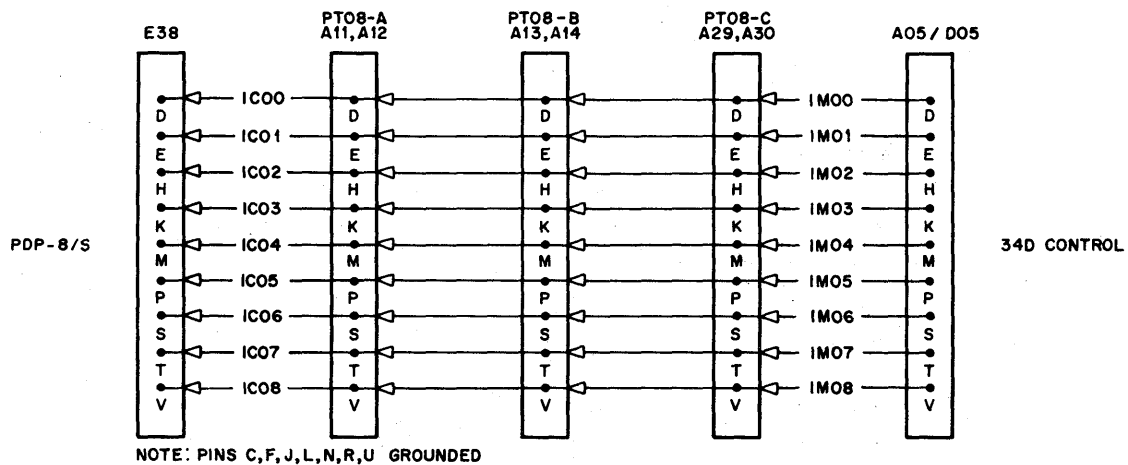


Figure 4-4 PDP-8/S to 34D Control Logic Interface

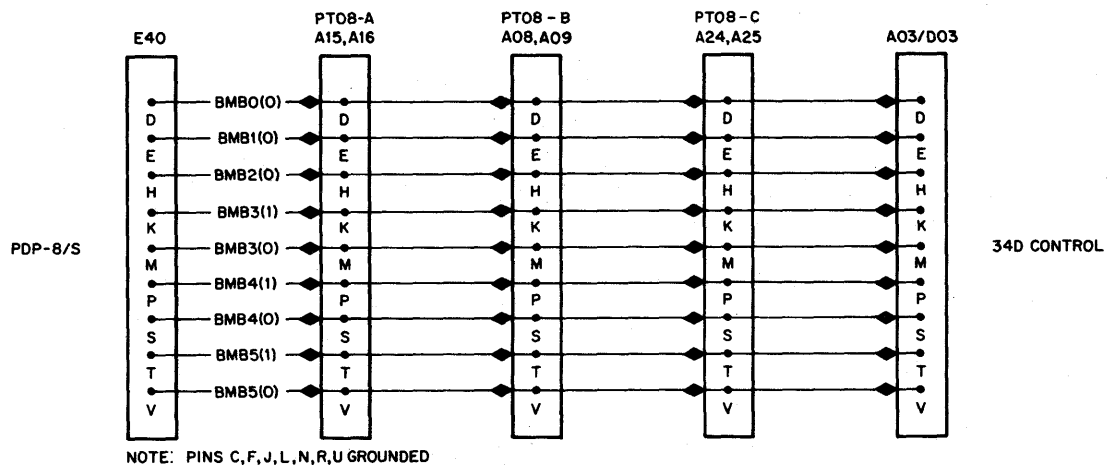


Figure 4-5 PDP-8/S Control Interface, Brightness, Light Pen, and X and Y Coordinate Control Address

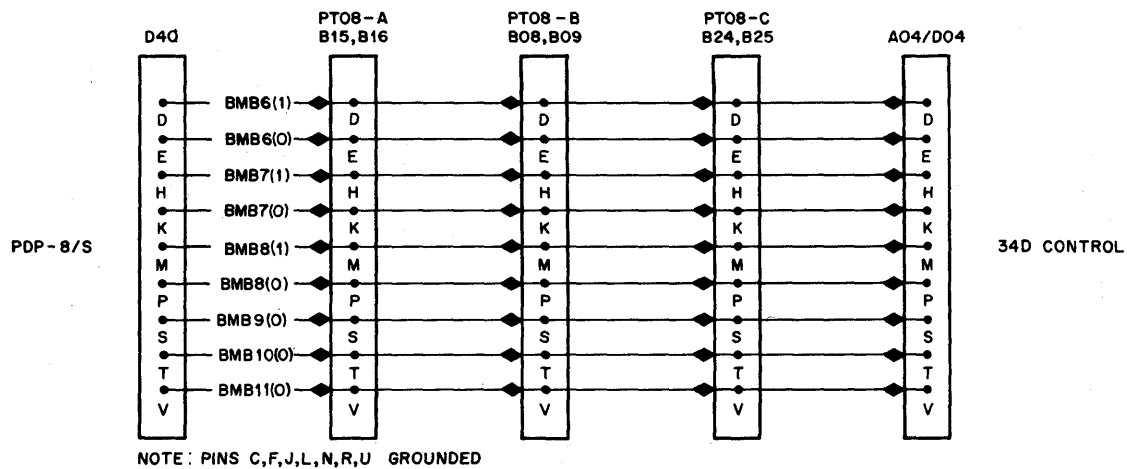


Figure 4-5 PDP-8/S Control Interface, Brightness, Light Pen, and X and Y Coordinate Control Address (continued)

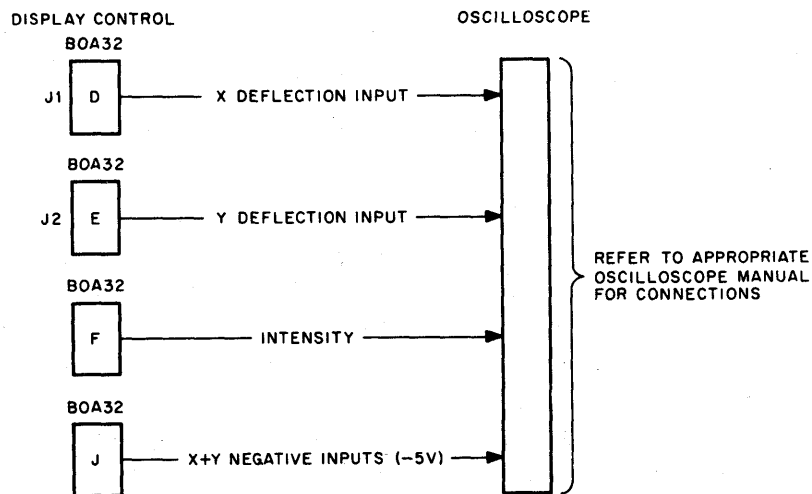


Figure 4-6 34D Control to Oscilloscope Interface

5. POWER SUPPLY

A DEC Type 779 Power Supply that is provided with the display control logic generates the three voltage levels (+10, -15, and -30 Vdc) required for operation of the display and its logic.

Section 8 of this document contains the schematic of the power supply DEC Drawing Number R5-B-779). Detailed information on the operational characteristics of this power supply is provided in the DEC Systems Modules Catalog (C-100).

6. MAINTENANCE

6.1 Preventive Maintenance

The general preventive maintenance procedures in the PDP-8 Maintenance Manual (DEC Doc. F-87) also apply to the display control logic.

6.1.1 Power Supply Checks - Table 6-1 lists the output voltage checks required for the DEC Type 779 Power Supply used in this equipment. To perform the power supply output checks described in Table 6-1, use a multimeter to make the output voltage measurements with the normal load connected, and an oscilloscope to measure the peak-to-peak ripple content on all dc outputs of the supply. The +10V and -15V supplies are not adjustable; therefore, if any voltage or ripple content is not within specifications, consider the power supply defective and initiate troubleshooting procedures.

Table 6-1 Type 779 Power Supply Output Checks

Measurement Terminals at Power Supply Output	Nominal Output (Vdc)	Acceptable Output Range (V)	Maximum Output Current (amp)	Maximum Peak-to-Peak Output Ripple (V)
Orange (+) to Yellow (-)	+10	+9.6 to 11.0	7.5	1.0
Yellow (+) to Blue (-)	-15	-14.5 to 16.0	8.0	0.4
Red (+) to Yellow (-)	+15	+14.5 to 16.0	7.5	1.1
Yellow (+) to Green (-)	+15	+14.5 to 16.0	7.5	1.1

6.2 Oscilloscope Maintenance

For information on maintenance and troubleshooting procedures, refer to the documentation provided by the manufacturer of the oscilloscope. The basic 34D Oscilloscope Display Unit utilizes a Tektronix RM - 503 Oscilloscope; however, other models may be used.

7 INSTALLATION

7.1 Cabinet and Desk Top Configuration

When the oscilloscope is supplied as either part of a cabinet-mounted or desk top PDP-8 system configuration, it normally mounts in a standard 19-in. radio rack adjacent to the central processor.

The logic controls and power supply for the Type 34D Oscilloscope Display Unit are mounted in the same cabinet as the oscilloscope when supplied as part of a desk-top configuration. Otherwise the logic controls and power supply are normally mounted in the PDP-8 cabinet itself.

7.2 Environmental and Power Requirements

The environmental and power requirements for the 34D logic controls are as specified in the PDP-8 Maintenance Manual (DEC Doc. F-87). For the oscilloscope specifications, refer to the appropriate oscilloscope manufacturer's documentation supplied with the Type 34D Oscilloscope Display Unit.

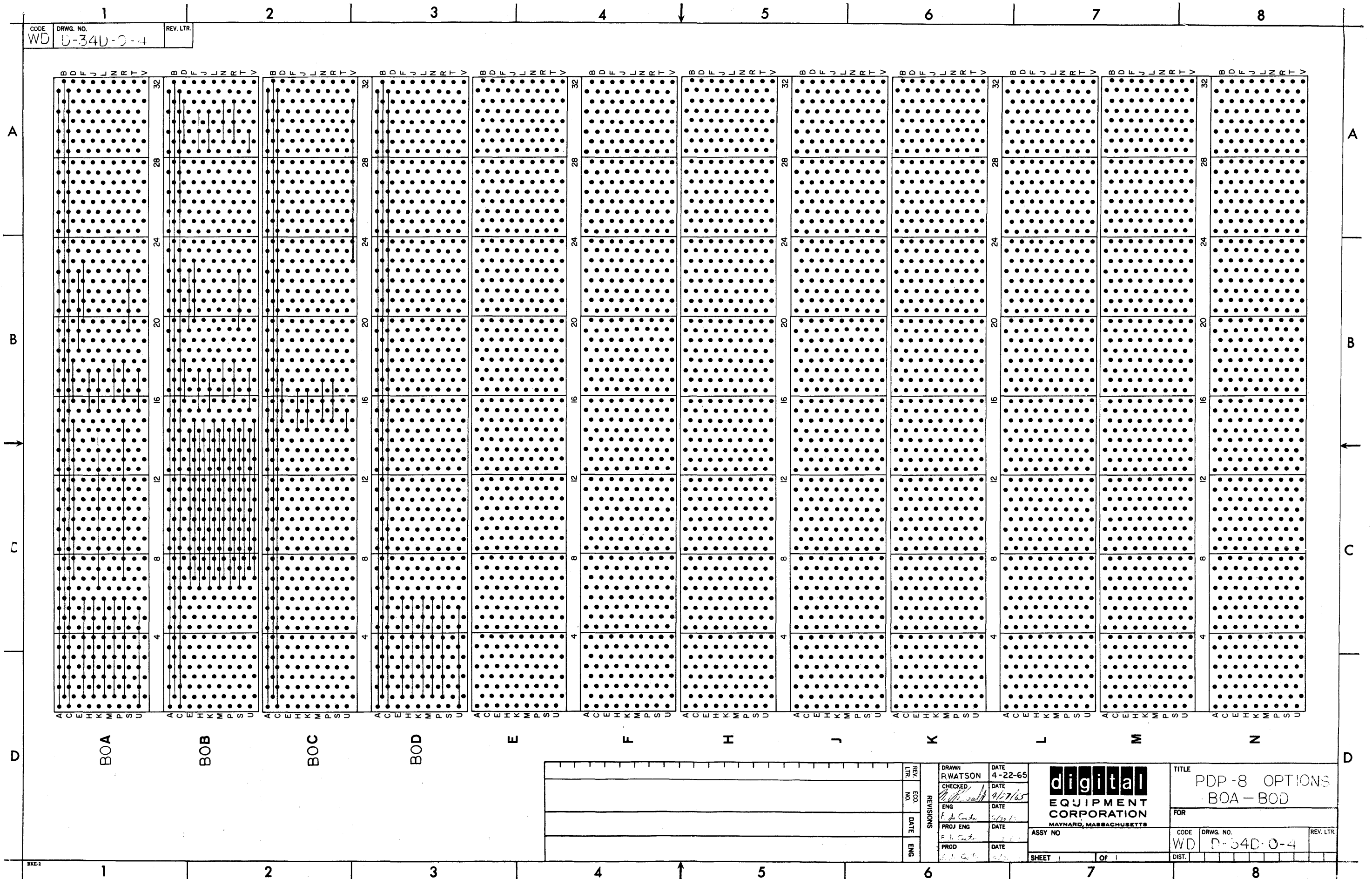
8. ENGINEERING DRAWINGS

This section contains the following logic diagrams, wiring diagrams, and module schematics which are pertinent to the Type 34D Oscilloscope Display Unit:

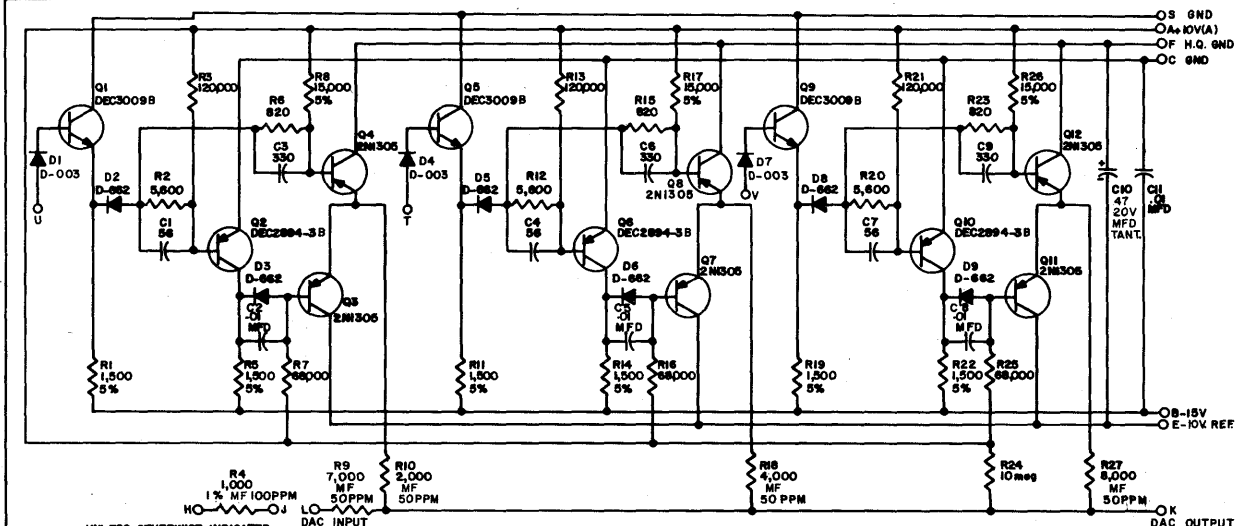
BS-D34D-0-2	Logic Diagrams
UML-D-34D-0-3	Utilization Module List
WD-D-34D-0-4	PDP-8 Options BOA-BOD

A601	} Module Schematics
A704	
R107	
R111	
R202	
R203	
R302	
W103	
W681	

[illegible]



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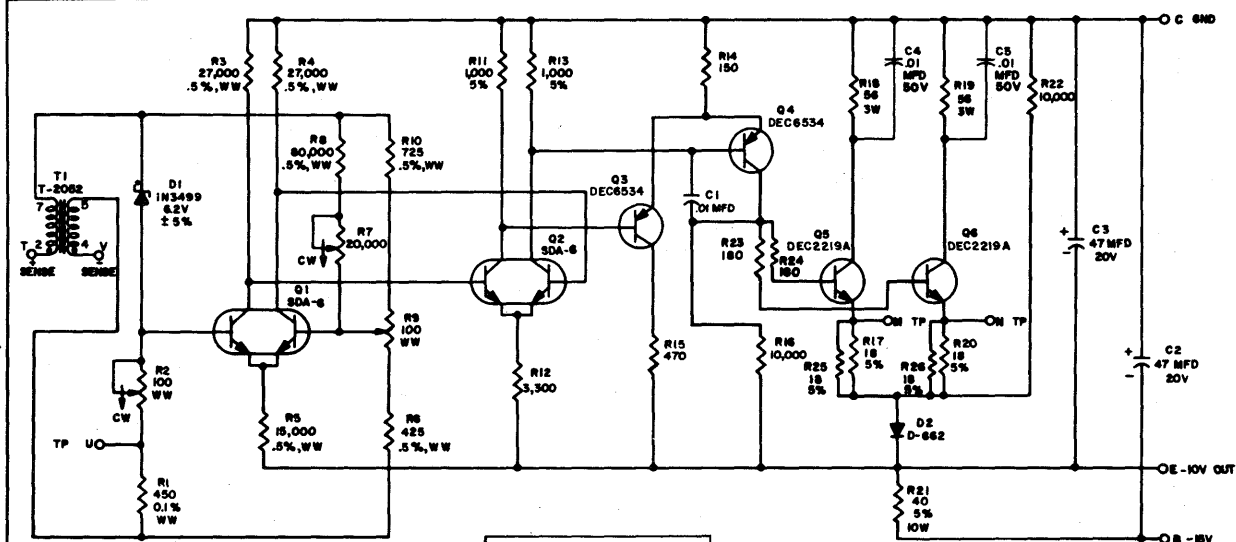
UNLESS OTHERWISE INDICATED:
RESISTORS ARE 1/4W; 10%
MF RESISTORS ARE 0.1W; 1/8W
CAPACITORS ARE MMFD

NOTE:
WHEN USED AS THE LEAST
SIGNIFICANT BIT, CONNECT J TO L
AND CONNECT H TO H.O. GND.

REVISIONS										DRN		DATE		TRANSISTOR & DIODE CONVERSION CHART				TITLE			
CHG NO. REV. DATE BY										W. PORTER		9-15-64		DEC		EIA		DEC		EIA	
1 4459 2 9-15-64 M.P.P.										CHK'D		9-30-64		2N3005		SAME		digital		3 BIT DAC A601	
2 4459 3 9-30-64 R.S.										ENG.		DATE		DEC2894-3B		NONE		EQUIPMENT		SIZE B	
3 4459 4 9-30-64 R.S.										R. SORESEN		9-30-64		DEC3009B		2N3009		CORPORATION		CODE CS	
4 4459 5 9-30-64 R.S.										PROD.		DATE		0003		1N84		MAYNARD, MASSACHUSETTS		NUMBER A601-0-1	
5 4459 6 9-30-64 R.S.														0662		1N845		PRINTED CIRCUIT REV.		REV. F	

DEC FORM NO. DRB 102

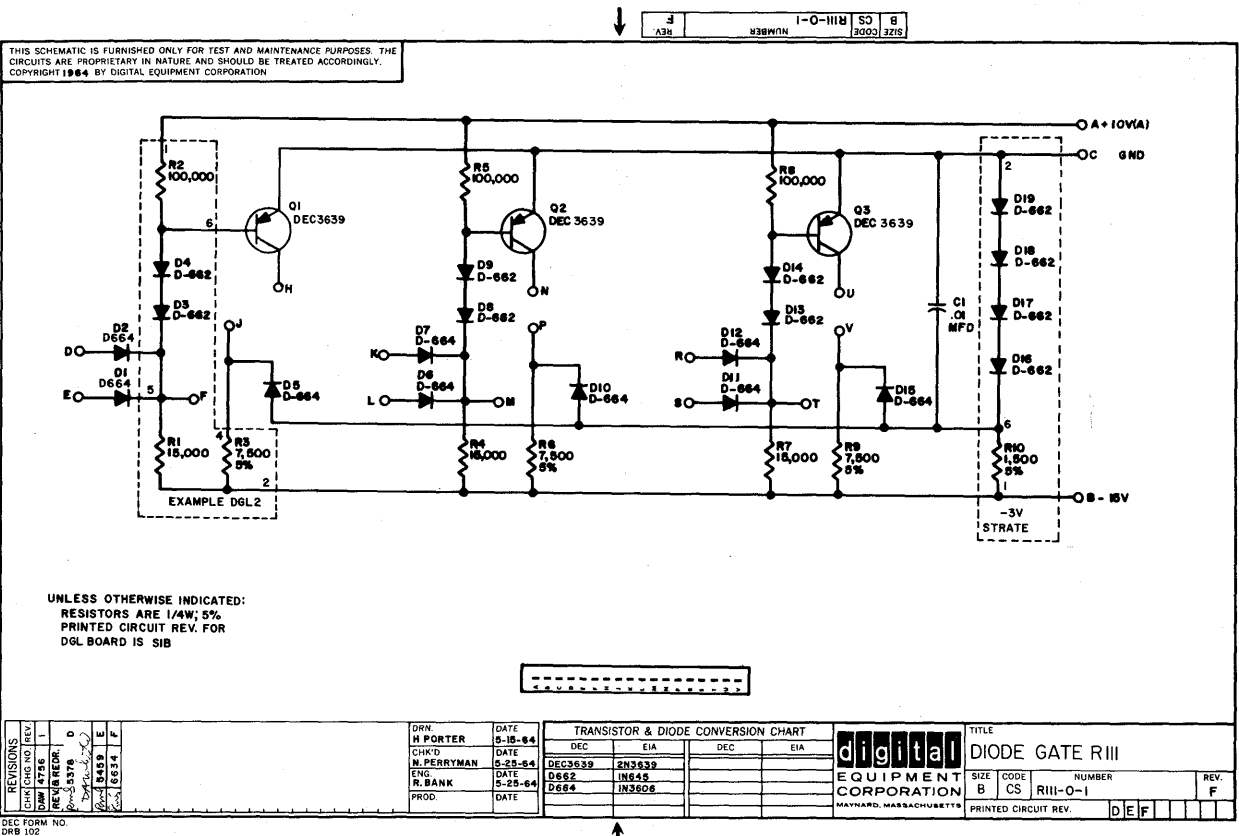
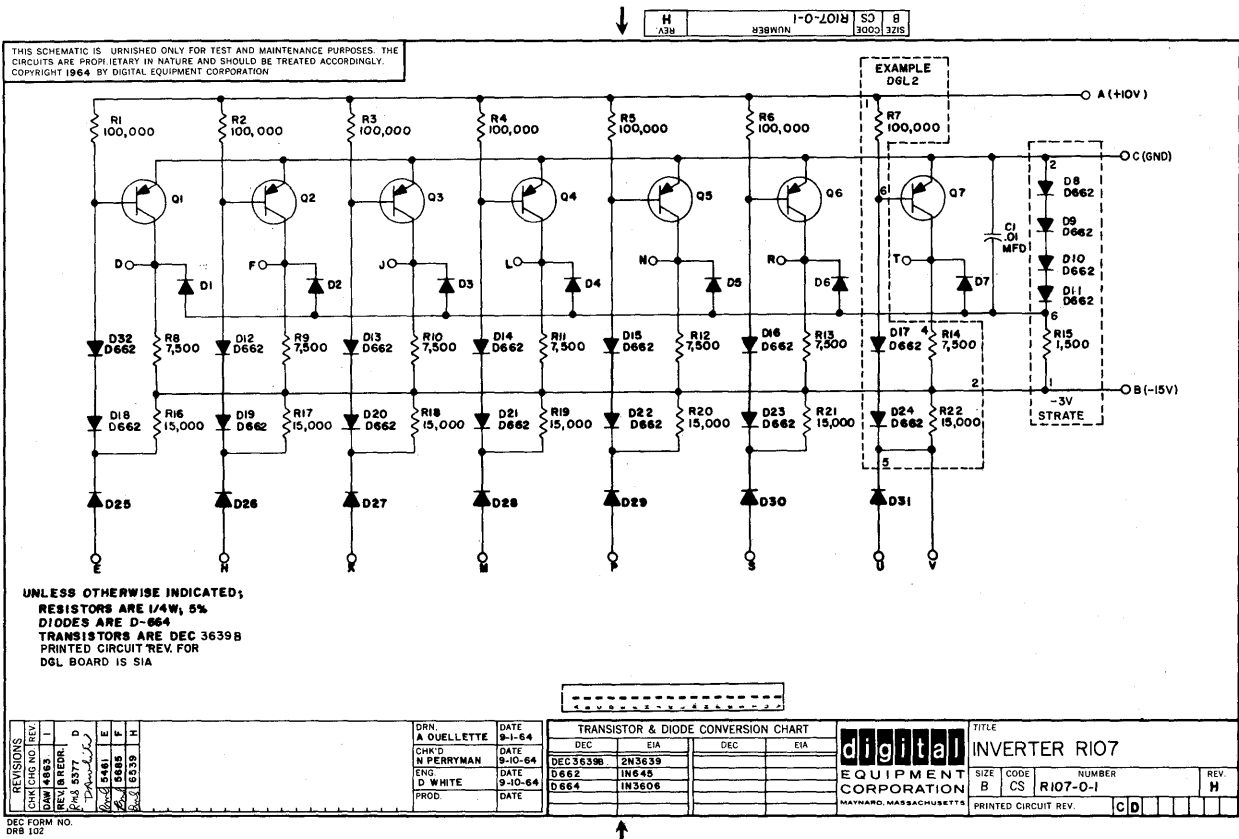
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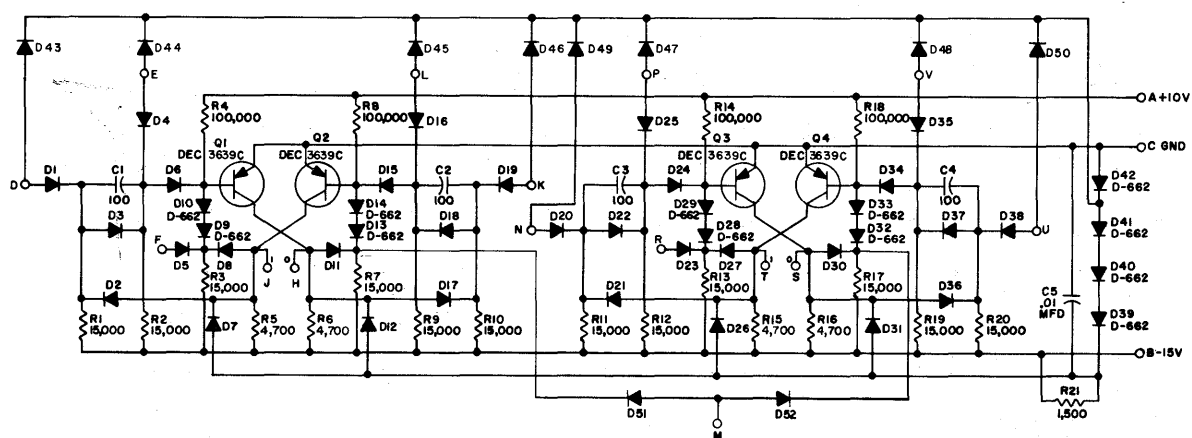
UNLESS OTHERWISE INDICATED:
RESISTORS ARE 1/4W; 10%
R10, R6 AND R1 ARE DAVEN 3PPM TYPE 1195
R3, R4, R5 AND R8 ARE DAVEN 20 PPM TYPE 1283
R2, R9 ARE 50 PPM DAYSTRUM TRANSITRIM
R7 IS A W-270P

H7 13 A 75107									
REVISIONS									
CHG NO.	REV.	DATE	BY	DRN	DATE	TRANSISTOR & DIODE CONVERSION CHART			
1	4535	1	9-3-64	A.O.	9-3-64	DEC			
2	4535	2	9-4-64	M.P.P.	9-4-64	EIA			
3	4535	3	9-4-64	R.A.G.	9-4-64	DEC			
4	4535	4	9-4-64	R.A.G.	9-4-64	EIA			
5	4535	5	9-4-64	R.A.G.	9-4-64	DEC			
6	4535	6	9-4-64	R.A.G.	9-4-64	EIA			
7	4535	7	9-4-64	R.A.G.	9-4-64	DEC			
8	4535	8	9-4-64	R.A.G.	9-4-64	EIA			
9	4535	9	9-4-64	R.A.G.	9-4-64	DEC			
10	4535	10	9-4-64	R.A.G.	9-4-64	EIA			
11	4535	11	9-4-64	R.A.G.	9-4-64	DEC			
12	4535	12	9-4-64	R.A.G.	9-4-64	EIA			
13	4535	13	9-4-64	R.A.G.	9-4-64	DEC			
14	4535	14	9-4-64	R.A.G.	9-4-64	EIA			
15	4535	15	9-4-64	R.A.G.	9-4-64	DEC			
16	4535	16	9-4-64	R.A.G.	9-4-64	EIA			
17	4535	17	9-4-64	R.A.G.	9-4-64	DEC			
18	4535	18	9-4-64	R.A.G.	9-4-64	EIA			
19	4535	19	9-4-64	R.A.G.	9-4-64	DEC			
20	4535	20	9-4-64	R.A.G.	9-4-64	EIA			
21	4535	21	9-4-64	R.A.G.	9-4-64	DEC			
22	4535	22	9-4-64	R.A.G.	9-4-64	EIA			
23	4535	23	9-4-64	R.A.G.	9-4-64	DEC			
24	4535	24	9-4-64	R.A.G.	9-4-64	EIA			
25	4535	25	9-4-64	R.A.G.	9-4-64	DEC			
26	4535	26	9-4-64	R.A.G.	9-4-64	EIA			
27	4535	27	9-4-64	R.A.G.	9-4-64	DEC			
28	4535	28	9-4-64	R.A.G.	9-4-64	EIA			
29	4535	29	9-4-64	R.A.G.	9-4-64	DEC			
30	4535	30	9-4-64	R.A.G.	9-4-64	EIA			
31	4535	31	9-4-64	R.A.G.	9-4-64	DEC			
32	4535	32	9-4-64	R.A.G.	9-4-64	EIA			
33	4535	33	9-4-64	R.A.G.	9-4-64	DEC			
34	4535	34	9-4-64	R.A.G.	9-4-64	EIA			
35	4535	35	9-4-64	R.A.G.	9-4-64	DEC			
36	4535	36	9-4-64	R.A.G.	9-4-64	EIA			
37	4535	37	9-4-64	R.A.G.	9-4-64	DEC			
38	4535	38	9-4-64	R.A.G.	9-4-64	EIA			
39	4535	39	9-4-64	R.A.G.	9-4-64	DEC			
40	4535	40	9-4-64	R.A.G.	9-4-64	EIA			
41	4535	41	9-4-64	R.A.G.	9-4-64	DEC			
42	4535	42	9-4-64	R.A.G.	9-4-64	EIA			
43	4535	43	9-4-64	R.A.G.	9-4-64	DEC			
44	4535	44	9-4-64	R.A.G.	9-4-64	EIA			
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46	4535	46	9-4-64	R.A.G.	9-4-64	EIA			
47	4535	47	9-4-64	R.A.G.	9-4-64	DEC			
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54	4535	54	9-4-64	R.A.G.	9-4-64	EIA			
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56	4535	56	9-4-64	R.A.G.	9-4-64	EIA			
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58	4535	58	9-4-64	R.A.G.	9-4-64	EIA			
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62	4535	62	9-4-64	R.A.G.	9-4-64	EIA			
63	4535	63	9-4-64	R.A.G.	9-4-64	DEC			
64	4535	64	9-4-64	R.A.G.	9-4-64	EIA			
65	4535	65	9-4-64	R.A.G.	9-4-64	DEC			
66	4535	66	9-4-64	R.A.G.	9-4-64	EIA			
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81	4535	81	9-4-64	R.A.G.	9-4-64	DEC			
82	4535	82	9-4-64	R.A.G.	9-4-64	EIA			
83	4535	83	9-4-64	R.A.G.	9-4-64	DEC			
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93	4535	93	9-4-64	R.A.G.	9-4-64	DEC			
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98	4535	98	9-4-64	R.A.G.	9-4-64	EIA			
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100	4535	100	9-4-64	R.A.G.	9-4-64	EIA			
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104	4535	104	9-4-64	R.A.G.	9-4-64	EIA			
105	4535	105	9-4-64	R.A.G.	9-4-64	DEC			
106	4535	106	9-4-64	R.A.G.	9-4-64	EIA			
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108	4535	108	9-4-64	R.A.G.	9-4-64	EIA			
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156	4535	156	9-4-64	R.A.G.	9-4-64	EIA			
157	4535	157	9-4-64	R.A.G.	9-4-64	DEC			
158	4535	158	9-4-64	R.A.G.	9-4-64	EIA			
159	4535	159	9-4-64	R.A.G.	9-4-64	DEC			
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161	4535	161	9-4-64	R.A.G.	9-4-64	DEC			
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182	4535	182	9-4-64	R.A.G.	9-4-64	EIA			
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186</									

DEC FORM NO. DRB 102



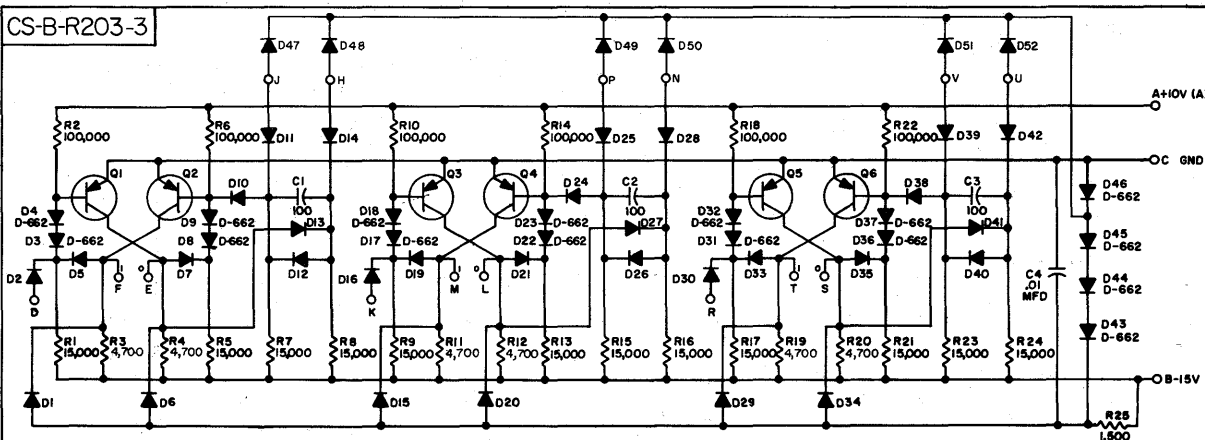
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UNLESS OTHERWISE INDICATED:
RESISTORS ARE 1/4W, 5%
CAPACITORS ARE MMFD
DIODES ARE D-664

REVISIONS CHG NO. REV. 4 DATE 6-23-64 BY 6849 F		DRN A. OUELLETTE DATE 6-16-64 CHK'D N. PERRYMAN DATE 6-18-64 ENG. R. BANK DATE 6-22-64 PROD. DATE		TRANSISTOR & DIODE CONVERSION CHART <table border="1"> <tr> <th>DEC</th> <th>EIA</th> </tr> <tr> <td>DEC 3639C</td> <td>2N3639</td> </tr> <tr> <td>D662</td> <td>1N646</td> </tr> <tr> <td>D664</td> <td>1N3608</td> </tr> </table>		DEC	EIA	DEC 3639C	2N3639	D662	1N646	D664	1N3608	digital EQUIPMENT CORPORATION MAYNARD, MASSACHUSETTS		TITLE DUAL FLIP FLOP R202 SIZE B CODE CS NUMBER R202-0-1 REV. F PRINTED CIRCUIT REV. DE	
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DEC 3639C	2N3639																
D662	1N646																
D664	1N3608																

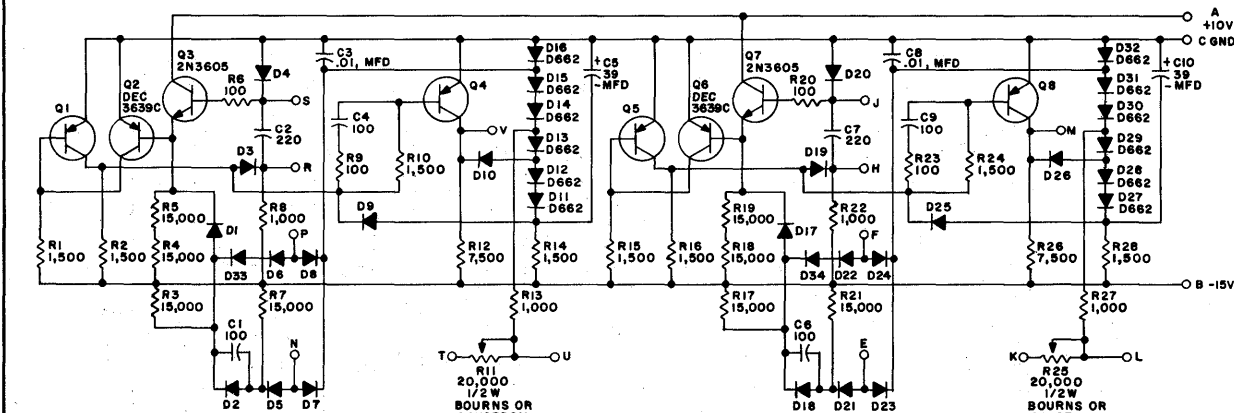
DEC FORM NO. DRB 102



UNLESS OTHERWISE INDICATED:
RESISTORS ARE 1/4W, 5%
CAPACITORS ARE MMFD
DIODES ARE D-664
TRANSISTORS ARE DEC 3639C

REVISIONS CHG NO. REV. 2 DATE 6-30-64 BY 6849 F		DRAFTSMAN A. OUELLETTE DATE 6-30-64 ENGINEER DATE 6-30-64 DATE		CHECKER DATE 7-6-64 PRODUCTION DATE		digital EQUIPMENT CORPORATION MAYNARD, MASSACHUSETTS		CODE CS DWG. NO. B-R203 REV. 3 TITLE TRIPLE FLIP-FLOP R203	
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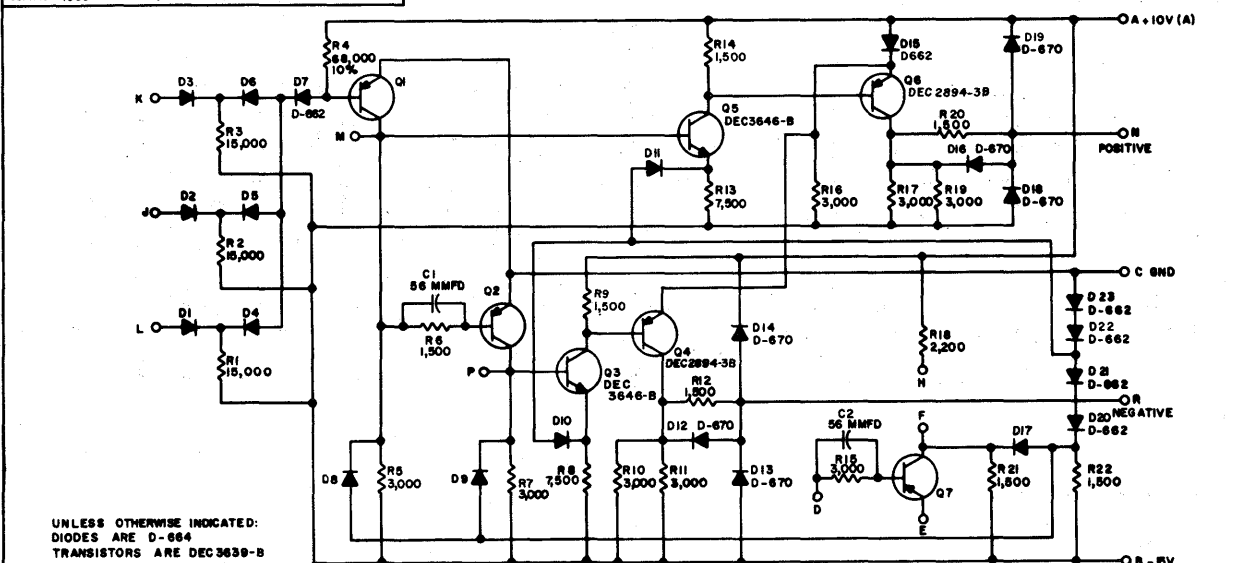


UNLESS OTHERWISE INDICATED:
RESISTORS ARE 1/4 W, 5%
CAPACITORS ARE MMFD
DIODES ARE D664
TRANSISTORS ARE DEC3639

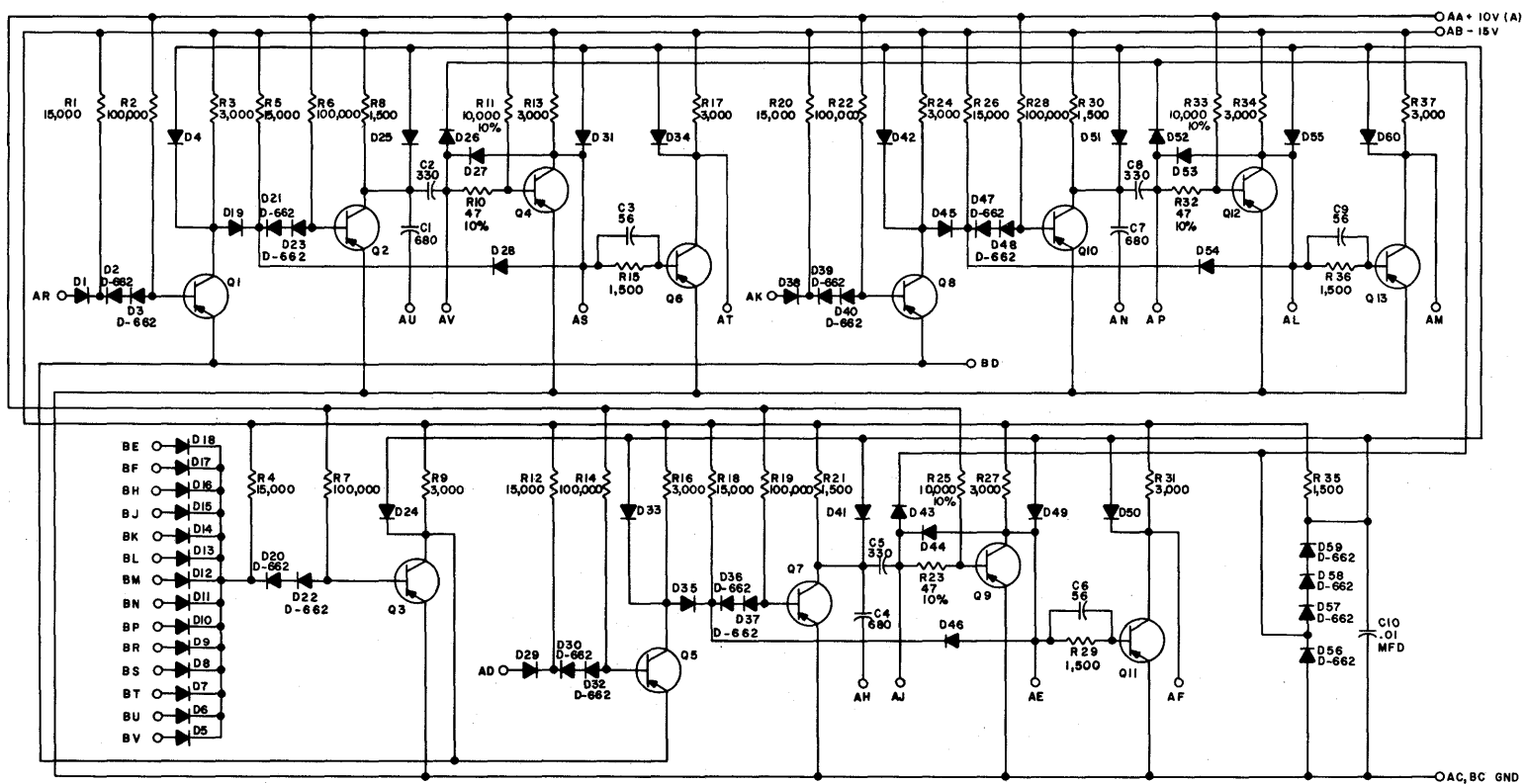
REV	CHG	NO	REV	DATE	DRN	DATE	TRANSISTOR & DIODE CONVERSION CHART	digital	TITLE	SIZE	CODE	NUMBER	REV.
1				8-12-64	A. QUELLETTE	8-12-64	DEC 3639 2N3639		DELAY R302	B	CS	R302-0-1	P
2				8-17-64	CHWD N. PERRYMAN	8-17-64	2N3605 SAME						
3				8-17-64	ENG. R. BANK	8-17-64	D662 IN645						
4					PROD.		D664 IN3605						
5							DEC3639C 2N3639						

DEC FORM NO. DRB 102

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UNLESS OTHERWISE INDICATED:
TRANSISTORS ARE DEC 3639
RESISTORS ARE 1/4 W, 5%
CAPACITORS ARE MMFD
DIODES ARE D-664



REVISIONS CHG NO. REV. DATE BY 1 1 1968 11/12 D 2 2 1968 11/12 D		DRN. I. NAHN DATE 4-21-68 CHK'D R. SILVERMAN DATE 4-22-68 ENG. R. SOBOLE DATE 4-22-68 PROD. DATE		TRANSISTOR & DIODE CONVERSION CHART <table border="1"> <tr> <th>DEC</th> <th>EIA</th> <th>DEC</th> <th>EIA</th> </tr> <tr> <td>DEC3639</td> <td>2N3639</td> <td></td> <td></td> </tr> <tr> <td>D662</td> <td>1N645</td> <td></td> <td></td> </tr> <tr> <td>D664</td> <td>1N3606</td> <td></td> <td></td> </tr> </table>		DEC	EIA	DEC	EIA	DEC3639	2N3639			D662	1N645			D664	1N3606			digital EQUIPMENT CORPORATION MAYNARD, MASSACHUSETTS		DEVICE SELECTOR W103 SIZE CODE NUMBER REV. C CS W103-O-1 D PRINTED CIRCUIT REV. C	
DEC	EIA	DEC	EIA																						
DEC3639	2N3639																								
D662	1N645																								
D664	1N3606																								

DEC FORM NO. DRC 102